

Freshwater ponds create highly dynamic Arctic tundra landscapes

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Abstract

Greenhouse gas emissions from Arctic tundra ponds and permafrost thaw provide important positive feedbacks to global warming. However, a high landscape heterogeneity and small size of ponds make it challenging to assess trends in surface water extent and associated carbon and energy fluxes, especially in the understudied Eastern Siberian tundra. Here, we show that surface water extent in these landscapes can be highly dynamic, shaped by pond-scale processes that cannot be detected in satellite products. Using a time-series of aerial imagery at 12 cm resolution spanning eight years (2014–2021), we classified surface water at three sites in Kytalyk National Park and traced the 465 ponds larger than 1 m². The surface water extent varied between 102%–124% relative to the time-series mean, without significant trends in contrast to previous reports. Individual pond area fluctuated by 52% on average, and two thirds of ponds were present for less than six years. One-quarter of ponds showed evidence for thermokarst or vegetation colonisation as drivers of change, based on our high-resolution surface elevation models. These findings highlight a highly dynamic nature of tundra ponds and stress the need for improved change detection to better model carbon and energy fluxes in this biome.

Main Text

Introduction

Greenhouse gas emissions from tundra ponds and permafrost thaw (thermokarst) provide important feedbacks to global warming (Wik et al. 2016, Olefeldt 2016). Ponds are a common feature of many Arctic landscapes, where they provide critical habitat for breeding birds (Corbeil-Robitaille et al. 2024) and form an important source of methane emissions (Wik et al. 2016). Methane emissions from ponds are mainly driven by microbial activity (Del Giorgio and del Cole, 1998) which is thought to increase with warming (Negandhi et al., 2016; Bartosiewicz et al., 2019). While permafrost thaw itself may release large quantities of carbon (Nitzbon et al. 2020, Wang et al. 2024), it can also be a driver of pond formation, with newly formed ponds having potentially elevated emission rates (Peura et al. 2020, Prėskienis et al. 2024). However, large uncertainties remain in the accounting and modelling of carbon fluxes in Arctic wetlands, especially in relation to small ponds (Negandhi et al., 2013, Melton et al. 2013, Muster et al. 2012, Wik et al. 2016) which are difficult to map with data exceeding metre-scale resolution (Muster et al. 2012). Further, ponds are an important factor of tundra land surface albedo and energy fluxes during summer (Juszak et al., 2017, Langer et al., 2011). Studying tundra ponds and how they change at fine spatial scales is therefore a key step to better understand the potential feedbacks to global warming.

There is currently no agreement on the direction of change in surface water area in response to warming across the Arctic. Some research indicates that water bodies, including small ponds, are decreasing across the Arctic (Webb et al. 2022, Wik et al. 2016, Andresen and Loughheed 2015). Other studies highlight the large methodological uncertainties in large-scale analyses of Arctic surface water change (Olthof et al. 2023, Webb et al. 2023). Additionally, there is considerable regional variation in trends

(Webb and Liljedahl 2023), especially for small water bodies (Payette, 2004, Walter et al. 2006, Wolfe 2011, Magnússon et al. 2021). For instance, increases in area and number of ponds has been reported for Northeastern Siberia (Magnússon et al. 2021), while declines have been reported in Alaska (Andresen and Loughheed 2015). The variation in regional trends for large water bodies has been associated with large-scale patterns in the distribution of continuous vs. discontinuous permafrost (Webb and Liljedahl 2023), but too little data are available for small water bodies.

A high landscape heterogeneity and water bodies with dynamic extents make it difficult to detect surface water changes in the Arctic tundra (Muster et al. 2012, Webb and Liljedahl 2023), and thus estimate the associated carbon fluxes at landscape scales (Muster et al. 2012, Hinks et al. 2015, Treat et al. 2018). The geomorphological processes that shape Arctic tundra landscapes can be very complex (Lara et al. 2020) resulting in a highly heterogeneous land cover at metre to decametre scales (Muster et al. 2012, Lara et al. 2015, Morgenstern et al. 2013). Tundra ponds are an important contributor to this metre-scale heterogeneity (Muster et al. 2012), with pond development being driven by various fine-scale abiotic and biotic processes (Pienitz et al., 2008), including freeze-thaw cycles of ice wedges (Lachenbruch 1962), permafrost thaw that creates depressions (Rautio et al. 2011, Luoto and Seppälä 2003) and vegetation colonisation by sedges and mosses (Magnússon et al. 2020). Some of these processes may occur on sub-decadal timescales (Magnússon et al. 2020) and could, combined with fluctuations in the main drivers the water budget of the tundra (precipitation, evaporation and lateral-flow) (Helbig et al. 2013), potentially cause rapid changes in the surface water area of the ponds. High-resolution (below 4–5 m) and high frequency spatial data are therefore needed to map tundra ponds at the landscape scale (Muster et al. 2012), but these are often not available over long time frames (Webb and Liljedahl 2023).

Here, we assess the change in surface water extent at the landscape-scale and for individual ponds in the tundra of Northeastern Siberia. We also test for evidence of thermokarst and vegetation colonisation as drivers of pond area change. To this end, we collected time-series of high-resolution (12 cm) drone imagery and digital surface elevation models (DSMs) at three study sites in the Kytalyk National Park for the time period 2014 to 2021 (Fig. 1). Two study sites are in a drained thaw lake basin with high and medium surface water occurrence (Fig. 1a) and one site is on a Yedoma ridge with low surface water occurrence functioning as a control. Our study thus covers two of the major landscape types in the region (Nitzbon et al. 2020, Schirrmeister et al. 2012). Next, we classified surface water in each image using a simple threshold in the blue chromatic coordinate (BCC) informed by annotated training data, and identified all individual ponds that exceeded 1 m² and were present for at least 3 years (465 in total; 357, 108, 0 respectively for the “high”, “medium” and “low” surface water occurrence sites). We then assessed how surface water extent changed at the landscape- and pond-level throughout each time-series. Finally, we tested whether thermokarst and vegetation colonisation contributed to individual pond change, by estimating the drop or gain in surface elevation in the drone-derived DSMs associated with the change. Our study thus allows us to gain novel insights into the rapid nature of changes in surface water and their ecological drivers in the understudied tundra landscapes of Northeastern Siberia.

Results

Surface water extent in Northeastern Siberian tundra landscapes is highly dynamic

The proportion of surface water detected was highly dynamic in the landscapes where ponds occurred regularly (“medium” and “high”, Fig. 1a). In the landscape with “low” pond occurrence, we only detected surface water in 2017, with no ponds occurring otherwise. The year 2017 was a strong outlier year with exceptionally high surface water cover across all study sites (Fig. 1b). The time-series for the “medium” and “high” sites showed high surface water variability (CV: 124% and 102%) even when the outlier year was removed (CV: 42% and 25% respectively). None of the time-series showed a significant trend over time (with and without 2017, Table S6, S7, S8, S9, S10, S11). We also did not detect any correlations with climate variables (Supplementary Methods, Table S12), but the time-series of surface water extent for the “high” site indicates some correspondence with mean summer air temperature (June-August) and autumn precipitation in the preceding calendar year (September-November) (Fig. S1, Table S12). The outlier year 2017 corresponded to a cold summer and high snowfall at the beginning of the preceding cold season (Tei et al. 2018), but annual temperature and precipitation values did not appear unusual in the context of the previous decades (Fig. S2).

Most ponds were short-lived and many strongly fluctuated in size

The majority of ponds were intermittently present in our time-series and the average variability in individual pond area was high (Fig. 2). Across all study sites and ponds, approximately two-thirds of ponds (65%) were present for less than six years and only one-third of ponds (35%) was “continuously” present, i.e., surface water was detected for six or seven years (allowing for one year of failed detection). The proportion of ponds “continuously” present was similar between the “medium” (38%) and “high” (34%) sites (Fig. S3a). The large majority of ponds experienced strong fluctuations in surface area (Fig. 2b). Excluding the outlier year 2017, 92% of ponds had a standardised variability (coefficient of variation, CV) larger than 10% of their mean surface area. The mean CV of pond area change was 52% across all time-series, with comparable values for the “medium” (52%) and “high” (52%) sites (Fig. S3b). We observed only 5 out of 465 (1.1%) ponds that were “stable” in the sense that they persisted for six or more years and had a CV of less than 10%, all five were found at the “medium” site.

Thermokarst and vegetation colonisation as drivers of pond change

We detected evidence for thermokarst (Fig. 3) and vegetation colonisation (Fig. 4) as drivers of surface area change in 24% of the ponds at the “medium” and “high” sites. Thermokarst may lead to pond expansion or movement (e.g., Fig. 3a). We observed expansion or movement of 68 ponds (15% of all ponds) across both sites, defined by a mean drop in surface elevation that exceeded 0.1 m (Fig. 3b) averaged across the pond area gained between 2014 and 2024. This value serves as our threshold for

detecting subsidence due to thermokarst (Online Methods). The proportion of these ponds with evidence for thermokarst was higher at the “medium” site (36 ponds = 33%) than at the “high” site (32 ponds = 9%) (Fig. S4). Vegetation colonisation, e.g., incursion of the pond area by sedges or mosses, can result in pond shrinking or even complete infilling (e.g., Fig. 4a). We observed vegetation colonisation in 83 ponds (18%) across both sites, defined by a gain in surface elevation that exceeded 0.1 m (Fig. 4b) averaged across the pond area lost between 2014 and 2021. This value serves as our threshold for detecting surface elevation gain indicating vegetation colonisation. The proportion of ponds with evidence for vegetation colonisation was higher at the “medium” site (40 ponds = 37%) than at the “high” site (43 ponds = 12%) (Fig. S5).

Discussion

Our findings demonstrate that tundra thermokarst landscapes and ponds can have highly dynamic extents - even over short time-periods such as three or five years. Very few ponds (five) in the studied landscape were stable in area and position. In contrast, the majority of studied ponds were short-lived and showed strong fluctuations in size. We found evidence for thermokarst and vegetation colonisation as drivers of change in approximately one quarter of the studied ponds, but many ponds changed without either driver being detected. Together, these findings have considerable implications for how we monitor change, estimate carbon and energy fluxes, and understand drivers of change in surface water extent in tundra thermokarst landscapes.

High temporal surface water variability complicates detection of long-term trends

High inter-annual variability of surface water at scales of metres to decametres restricts our ability to derive estimates of long-term trends, such as wetting or drying, using remote sensing in tundra systems with pond presence. Multiple studies have already highlighted the insufficiencies of moderate resolution (10–250 m) satellite imagery for surface water (and trend) detection in such tundra systems (Muster 2012, Webb and Liljedahl 2023, Mullen et al. 2023). While centimetre-resolution drone data (as shown by us) and, to some degree, commercial high-resolution satellite imagery (Magnusson et al. 2020, Magnusson et al. 2021, Mullen et al. 2023) can provide data to detect surface area changes in small water bodies at the relevant spatial scales, our findings highlight the need to also sample these systems over long time-frames and at a high temporal frequencies (at least annual, likely better: intra-annual, see also Mullen et al. 2023). Even if an underlying long-term trend was present, time-series like ours that span almost a decade are not statistically powerful enough to detect a trend, given the high variability observed. We therefore suggest caution when interpreting trends based on time-series with moderate to low spatial (10–250 m) and temporal resolution (i.e., larger than annual) in those tundra systems where small ponds are present and encourage the continued development of improved monitoring methods (Webb and Liljedahl 2023, and Mullen et al. 2023).

Pond dynamics driven by more than just thermokarst and vegetation colonisation

Many ponds in our study showed high surface water variability without evidence for either thermokarst or vegetation colonisation. Pond-dominated tundra landscapes have been described as “fill-and-spill” systems (Helbig et al. 2013), where surface water dynamics are driven by snow melt (a function of winter precipitation) during spring and evaporation during summer, as well as surface and subsurface lateral flow (Woo and Guan 2006, Wolfe et al. 2011, Helbig et al. 2013). These processes are likely also key drivers of surface water dynamics observed in our study, as illustrated by the outlier year 2017: The year 2017 was marked by the inundation of landscapes across the region, that was driven by a high amount of snowmelt lasting into mid-summer, following a larger than unusual precipitation in the preceding October (Tei et al. 2020). Quantifying and modelling the relative contributions of the different drivers of landscape-level pond change (for example, using drone-derived surface models) will be critical for improving our predictions of future landscape dynamics and the associated carbon fluxes.

Surface elevation changes integrate vegetation change and permafrost processes

Our findings confirm that thermokarst and vegetation colonisation are processes contributing to pond change in the study region (Magnússon et al. 2020), but our method cannot resolve the complex causes of these processes themselves. At the “medium” and “high” study sites the ice-content of the first 60 cm of soil ranges from 50–70% (Wang et al. 2019). If such soils are thawed, the release of water will induce subsidence (Li et al. 2017) causing thermal erosion. By measuring the drop in surface elevation we detect this subsidence but we also integrate the loss of the vegetation canopy. Indeed, thermokarst is often associated with loss of vegetation, which can be both cause and effect of thermal erosion (Kanevsky et al. 2017, Li et al. 2017, Magnússon et al. 2020). Similarly a gain in surface elevation integrates vegetation growth in previously water covered areas and the formation of ice rich permafrost, which also can be cause and effect of each other (Liljedahl 2016). Deriving digital elevation and canopy height models using drone-based LiDAR or dense ground control points could enable the separation of these processes, but in the absence of these data our study is limited to measuring the integrated effects.

Studying fine-scale pond dynamics will better our understanding of ecosystem functioning

The strong variation in the surface area of individual ponds observed in our study suggests a need to better understand the associated consequences for the ecosystem and carbon and energy cycle. Many ponds in our study fluctuated notably in size and many had a short life-span; temporary flooding (Woo and Guan 2006) and drying out (Smol and Douglas 2007) were common across the time-series, as was

the establishment and breakage of surface water connections between ponds (Wrona et al. 2016). How these dynamic processes affect the habitat availability for breeding birds (Corbeil-Robitaille et al. 2024) or the assembly of microbial communities (Le Moigne et al. 2020, Gurney et al. 2022) and their impact on the functioning of the ponds in the carbon cycle or other landscape-scale processes (Blackburn-Desbiens et al. 2023) are not well understood. Furthermore, processes like thermokarst and vegetation colonisation have different implications for the carbon cycle. Thermokarst will likely release old carbon from the thawed soil (Wauthy et al. 2018), while vegetation incursion by sedges will act as a conduit for the release of methane from the pond water (Knoblauch et al. 2015). Combining time-series of high-resolution imagery with in-situ field studies of macro fauna or microbial dynamics could therefore greatly advance our understanding of the effects of pond dynamics on landscape-scale ecosystem function.

Not all tundra landscapes experience directional trends or stability

Overall, our study highlights the possibility that not all tundra systems are experiencing clear directional trends or stable states. Instead, our data shows that tundra landscapes with a high amount of pond presence can be very dynamic in space and time (Helbig et al. 2013). Those dynamics are likely to increase further as extreme weather events become more common while the tundra continues to warm (Wolfe et al. 2011, Tei et al. 2020). Considering the dynamic nature of these systems is critical when assessing long-term trends in surface cover of tundra landscapes, including surface water change (e.g., Webb et al 2022, Olthof et al. 2023), and their associated impacts on ecosystem function. The dynamic ponds further contribute to the complexity of greening and browning (Myers-Smith et al. 2020, Beamish et al. 2020) and albedo trends (Plekhanova et al., 2022) in medium- to low-resolution satellite data. Only by accounting for the dynamic nature of some tundra landscapes will we be able to successfully estimate these trends and model the carbon and energy budget in the Arctic.

Online methods

Study sites

The three study sites are located in Kytalyk National Park in Northeastern Siberia, Russia in proximity to the Chokurdakh Research Station (70.83°N, 147.49°E, Interact: <https://eu-interact.org/field-sites/chokurdakh-scientific-tundra-station/>).

The climate in the region is cold, with an average annual temperature of -13.4 °C. Summers are mild with an average July temperature 10.3 °C. The mean annual precipitation is low (196 mm) with the majority falling in autumn and winter as snow, approximately 39% of the precipitation falling over summer (June-August). These climate data are based on the Chokurdakh weather station, WMO 21946, 28 km southeast of the Chokurdakh Research Station and averages refer to the period between 1981 and 2010 (Nauta et al. 2015).

The landscapes in the region are underlain by Ice rich permafrost with an average active layer thickness of 42 cm and the soils are composed of silt with a shallow upper layer of peat (Grysko et al. 2021). The most common landscape types in the region include drained thaw lake beds (Alas depressions) and Yedoma ridges (Nitzbon et al. 2020). Ponds and patterned ground (ice-wege polygons) are common in the drained thaw beds (Nitzbon et al. 2020), where the vegetation is composed of dwarf shrubs (*Betula nana*), moss tundra and sedges (Iturrate-Garcia et al. 2016). Ponds are rare on the Yedoma ridges (Nitzbon et al. 2020), where the vegetation primarily consists of tussock sedges (*Eriophorum vaginatum*) (Iturrate-Garcia et al. 2016). Holocene deposits, thermokarst lakes (not drained) and river channels are also present in the region (Nitzbon et al. 2020), but we did not consider these in our study.

Within the constraints of the drone data available, we chose the three study sites with an average dimension of 475 m x 435 m to capture the range of pond density occurring in the landscapes of the region. We selected a section of a drained thaw lake bed with a high pond occurrence (site “high”, area: 19.75 ha, centre: 70.8322°N 147.4779°E); a section of a drained thaw lake bed with a medium pond occurrence (site “medium”; area: 21.67 ha; centre: 70.8336°N 147.4951°E); and a section of a Yedoma ridge with no permanent pond occurrence as a control site (site “low”, area: 16.36 ha, centre: 70.8292°N 147.4638°E) (Fig. 1a). For further details see Supplementary Methods - “Determination of site boundaries” and Table S1.

Drone data collection

We collected Red-Green-Blue (RGB / true colour) drone imagery at a near annual basis from 2014 to 2021. For a detailed description of all flights see Table S2. In brief, we used eBee Classic and eBee X PPK drone systems (AgEagle, formerly SenseFly, Lausanne, Switzerland) equipped with either a SenseFly S110 RGB (AgEagle) or SenseFly S.O.D.A (AgEagle) sensor. Average ground sampling distances varied between 2.00 cm and 4.36 cm based on sensor and flight altitude (Table S2). We carried out all flights during summer between 11 July and 18 August of a given year. Repeat surveys within a year were available from 2019 for the “medium” (3x) and “high” sites (2x). No data was collected in 2015 and the 2016 data at the “low” site was of insufficient quality. In some instances, a merging of multiple flights within a year was required to achieve complete coverage for a site. Illumination conditions varied widely (sunny, overcast, time-of day, etc.) (Table S2).

Drone data processing, geolocation and colour correction

We used Pix4D Mapper (Pix4D S.A. Prilly, Switzerland) to generate and co-register the RGB raster mosaics and surface models (DSMs) for each drone survey.

We generated the RGB mosaics and surface models using the “3D Maps” template in Pix4D and set the output resolution to 12 cm to obtain the mosaics and DSM with a uniform ground sampling distance (GSD). This distance is approximately three-times the maximum GSD in our dataset (Table S2). In four

cases, the illumination of the individual raw images varied strongly, resulting in patchy mosaics. We addressed this issue using the *colorCorrection* function of the *CRImage* R package (Failmezger et al. 2024) to standardise the colour profiles of the individual images within the four mosaics prior processing (Table S2).

High-precision geo-location using dGNSS post processing kinematics (PPK) was only possible for the 2020 and 2021 surveys using the newer eBee X systems. We therefore chose the 2021 mosaics at each site as the reference mosaic and co-registered all preceding mosaics in the time-series using manual tie-points in Pix4D. We verified the geolocation accuracy relative to the 2021 reference using five independent check points. The resulting estimate of the mean geolocation accuracy did not exceed 46 cm (three-times the 12 cm resolution of the mosaics) in all cases (Fig. S6). For further details see Supplementary Methods “Geolocation and accuracy assessment”.

Finally, we standardised the colour profiles of all RGB drone mosaics relative to the mosaic from the 2017 survey at the “high” site using the *colorCorrection* function of the *CRImage* R package (Failmezger et al. 2024) to aid with the next surface water detection steps. A calibration using empirical lines (Wang and Myint 2015) was not possible as standardised targets were not deployed during the drone surveys, instead we used a surface water detection approach that worked on a similar empirical principle using a mosaic specific threshold based on annotated training data.

Surface water detection

We used a simple threshold in the Blue Chromatic Coordinate (BCC) determined from supervised training data to classify surface water in each RGB mosaic. We calculated the BCC as the ratio between the digital number (DN) of the blue band and the sum of all bands ($BCC = DN_{blue} / (DN_{red} + DN_{green} + DN_{blue})$). We then produced a set of manually annotated reference polygons with two cover classes (“surface water” and “other”) for each drone mosaic (Table S3) and extracted the BCC for all pixels in these polygons. Next, we iterated through all potential BCC threshold values from 0 to 1 at 0.01 intervals and determined the classification accuracy for each threshold to separate surface water from other surfaces based on the annotated reference data. We then chose the threshold providing the highest overall classification accuracy for each drone mosaic (Table S4) and generated a corresponding raster file with two surface classes (“surface water” and “other”). For further details on training and classification see Supplementary Methods - Training annotations & accuracy assessment. Finally, we removed all clusters of surface water pixels less than 1 m², which was the size of the smallest annotated pond in our training data. The mean pond size was 17.89 m².

Analysis of landscape-scale dynamics in surface water extent

To assess the dynamics of surface water extent at the landscape-scale, we calculated the area of surface water for each site and survey-time combination, and then analysed its variability within the three annual time-series. To this end, we determined the coefficient of variation (CV, the standard deviation divided by the mean and multiplied by 100%) for each time-series, once with and once without the outlier year 2017. For those year and site combinations for which repeat surveys were available (2019 and “high” and “medium” sites) we used the mean value of all surveys for the given year when calculating the CV. Based on the few available repeat surveys, the variation in surface water area extent within 2019 relative to the mean of the whole time series ($sd_{2019} / mean_{all} \times 100$) was small compared to coefficient of variation across the whole time series ($sd_{all} / mean_{all} \times 100$): 3% (2019) vs 124% (all years) for the “high” site and 15% (2019) vs 102% (all years) for the “low” site. See Table S5 for a detailed listing of all surface water area and proportion values. We also tested for trends in all time-series using OLS and correlations with climate variables (Supplementary Materials - Surface water trend and climate analysis).

Analysis of dynamics in individual pond area

To assess the dynamics in surface water extent at the pond-scale, we identified all individual ponds in each time-series and then analysed the change in their surface water area.

We identified the individual ponds by generating a binary composite of all surface water extent maps for each time-series. In this composite, a cell was set to one (water present) if water was detected in this cell at any point in the time-series. All other cells were set to NA. In this component of the analysis, we excluded the extreme outlier year 2017 as many ponds were connected during this year. We also excluded all repeat surveys within a year and only kept one observation per year. Here we excluded all surveys that were furthest from the median within-year timing of observations across all years (mosaic IDs: “cbh_2019”, “tlb_2019_a” and “tlb_2019_b”). We then obtained the polygon geometries of all clusters of surface water in the composite (hereafter “reference ponds”) and tracked their presence across the time-series.

We tracked the reference ponds identified from the composite across the time-series by testing for geometric intersection with all ponds detected in a given year. As surface water at the “low” site was only detected in 2017, we removed this site from all subsequent analysis. Next, we assigned each remaining cluster of surface water an individual pond identifier and determined the number of years the pond was present in the time-series by testing for geometric intersection between polygons of the clusters derived from the composite with the polygons of all clusters of water detected in each year. To exclude temporary “puddles”, we kept only those clusters of surface water that were present at least three times in each time-series. Experimental manipulation of ponds had previously been carried out at the “medium” site; we identified and removed these ponds (total: 8) from all subsequent analyses. Finally, we calculated the pond area in each year as the total area of all clusters of surface water

intersecting with the pond geometry derived from the composite (some ponds split), and then determined the CV as a measure of pond area variability, excluding the outlier year 2017.

Analysis of drivers of pond-change

We used the drone derived digital surface model to identify the presence of thermokarst and vegetation colonisation as drivers of pond surface area change. We restricted our analysis to changes in the relative surface elevation in the surroundings of each pond, as we were unable to precisely geolocate the three-dimensional point clouds in the absence of high-precision geolocation of image tags and ground control points. To this end, we cropped the drone DSMs for each year with a buffer area of 10 m around each pond and masked all areas that were detected as water in any point of the time-series (including 2017). We then standardised the DSMs relative to the minimum surface elevation within this area.

For the detection of thermokarst as a driver of pond change, we calculated the mean drop in surface elevation in the DSMs between 2021 and 2014 within the pond area gained (Fig. S7a). For the detection of vegetation colonisation as a driver of pond change, we calculated the mean gain in surface elevation in the DSMs between 2021 and 2014 for the pond area lost (Fig. S7b). We calculated the average drop / gain in surface elevation for the pond area gained / lost rather than the total absolute change in volume as the ponds varied strongly in size. In other words, we standardised the volumetric differences by area gained / lost to obtain a comparable measure of change in surface volume across ponds independent of size.

It is important to highlight two key contributors to uncertainty in the above-described method. First, standardising the DSMs by the minimum elevation within the buffered surroundings of each pond assumes that both the minimum elevation as well as the measurement error are constant across the time-series space. The assumption will likely not hold true in all cases and may introduce both random and systematic errors. Second, for the calculations of the volume differences we assumed that any area covered by water has a relative surface height of 0 m. This assumption may lead to a systematic under- or overestimation of the volume lost in the thermokarst and vegetation colonisation detection analyses respectively. Considering all of the above, we estimated that we are only able to reliably detect volume changes in the surface model that exceed $0.1 \text{ m}^3 \text{ per m}^2$, which is equivalent to a drop (or gain) in the surface elevation of 10 cm. Hence, we only assigned the detection of thermokarst or vegetation colonisation to a pond where the threshold was exceeded in the time-series. We empirically validated this threshold by manually inspecting the RGB / DSM time-series of all ponds.

Software

We conducted all data processing and statistical analysis in R ver 4.4.0 (R Core Team 2024). We used the sf package ver 1.0-16 (Pebesma and Bivand 2023, Pebesma 2018) for all geometrical operations; the terra package ver 1.7-78 (Hijmans 2024) for raster handling and calculations; the tidyverse ver 2.0.0.0

(Whickam et al. 2019) and pbapply ver 1.7-2 (Solymos and Zawdazki 2023) packages for general data wrangling and parallel processing; the ggplot2 ver 3.5.1 (Wickham 2016), cowplot ver 1.1.3 (Wilke 2024) and tidyterra ver 0.61 (Hernangómez 2023) packages for generating the maps and figures, and the gt package ver 0.11.0 (Iannone et al. 2024) for generating tables.

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Declarations

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Code and Data availability

All code, tabular and geometric data required to regenerate the analyses are available via the following GitHub repository: https://github.com/jakobjassmann/pond_project

The raster data (RGB drone mosaics and DSMs) are also required and are available on Zenodo: <https://doi.org/10.5281/zenodo.13992090>

The drone RGB rasters, surface water classifications and individual pond time-series have been visualised in a Web-GIS hosted on AWS: <https://pondproject.s3.eu-central-1.amazonaws.com/index.html>

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Figures

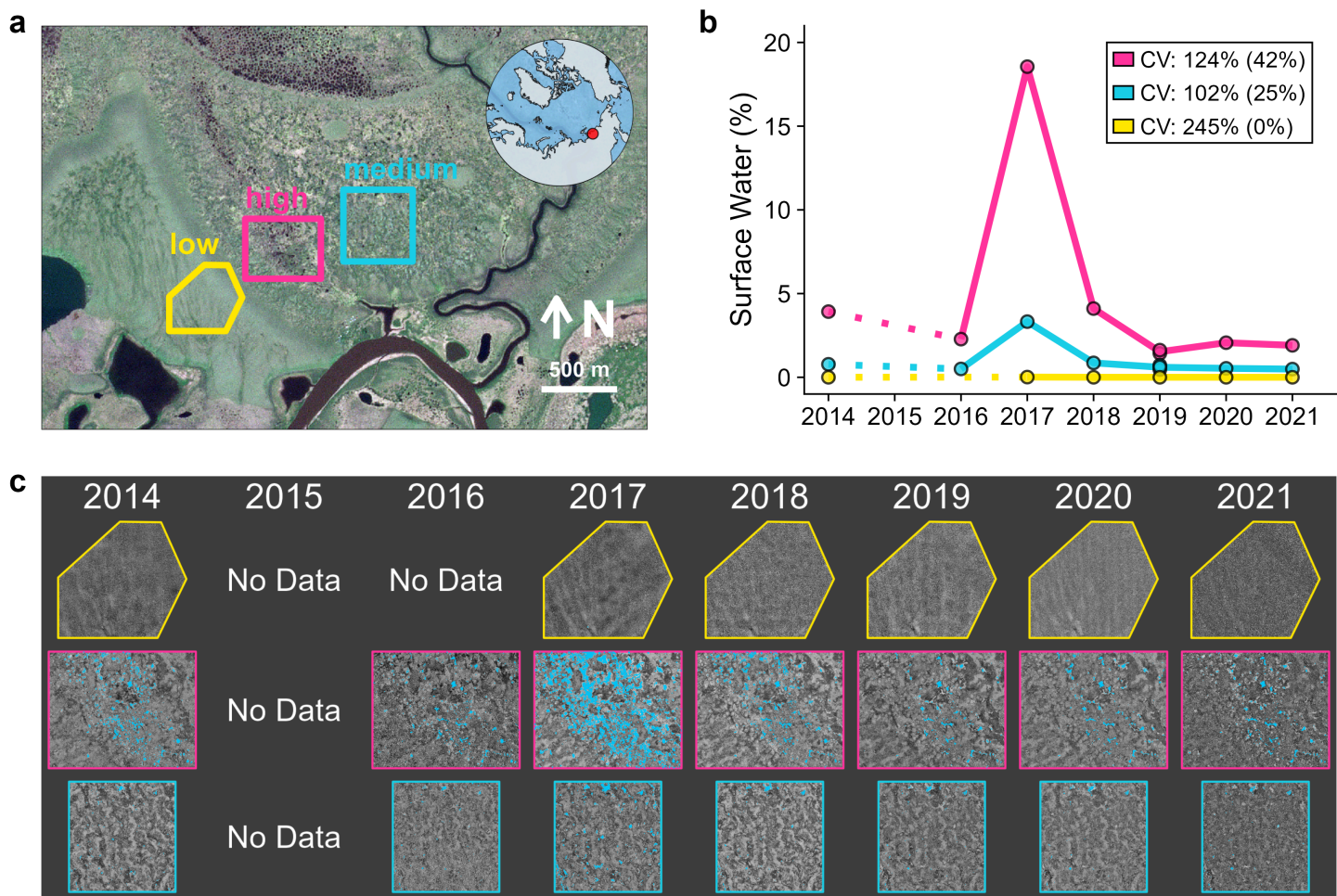


Figure 1

Surface water extent in Northeastern Siberian landscapes can be highly dynamic. (a) Map showing the locations of the three Arctic tundra landscapes with high, medium and (very) low pond occurrences in Northeastern Siberia. (b) Time-series of surface water proportion for each site based on classifications of true colour drone-imagery indicating high variability. Dashed lines in the graph indicate gaps where no drone data was available. Coefficients of variation (CV) for each time-series are given in insert on the top-right, values in parentheses represent the CV with outlier year 2017 removed. (c) Surface water maps for each study site, showing the surface water (light blue) on top of the desaturated true-colour drone imagery. No data was collected in 2015 across all sites, and the data from 2016 for the “low” site was of

insufficient quality for accurate geolocation. Satellite Imagery in (a): © Planet Labs 2022 (Planet Labs PBC 2022).

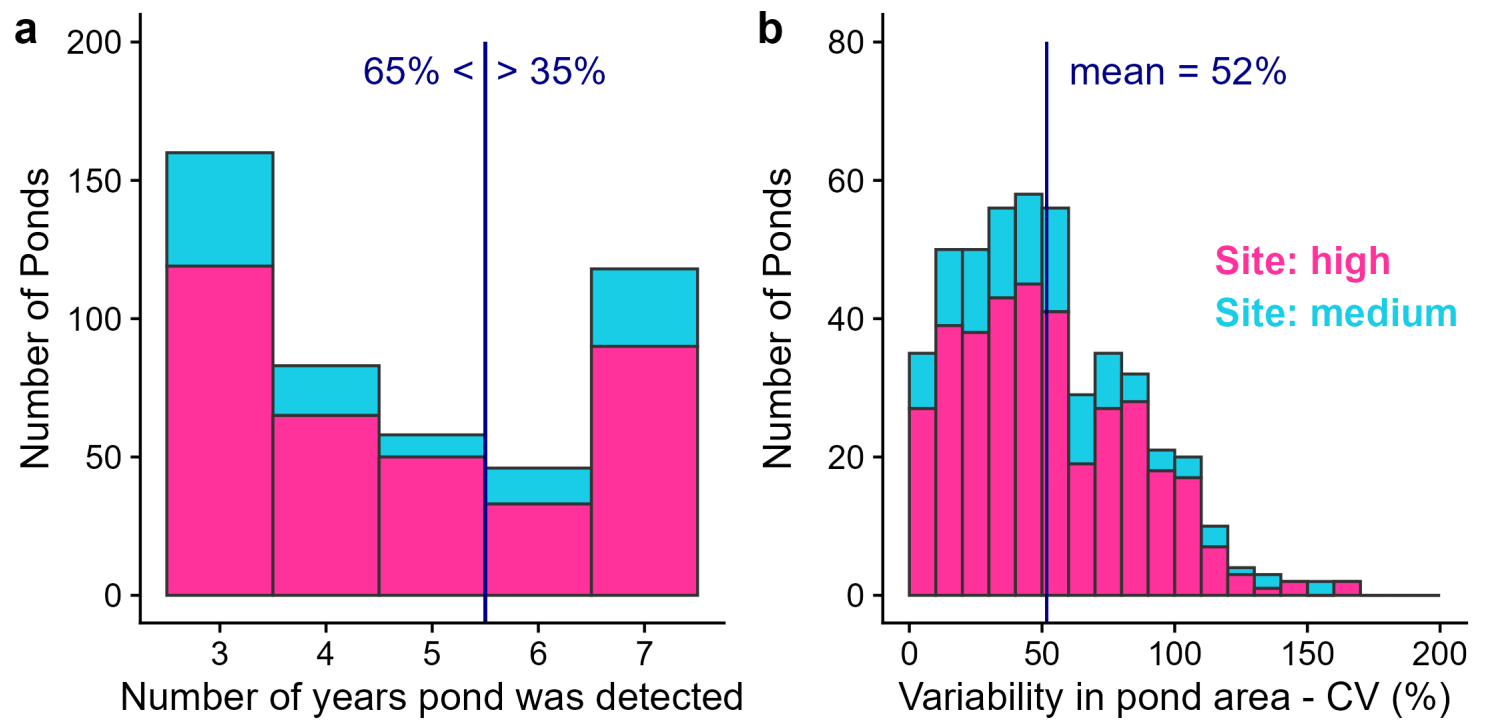


Figure 2

Ponds were short-lived and fluctuated strongly in size. The majority of ponds occurred for less than six years in the time-series and individual ponds showed a high variability in their surface area across the time-series. (a) Distribution of the number of years each pond was detected in the time-series. The dark blue line indicates the proportional split between those ponds present less than six years and those present “continuously” for six or more years (allowing for one year of failed detection). (b) Distribution of the coefficient of variation (CV) of the pond’s area across each time-series. The dark blue line indicates the mean CV value for all ponds. CV values were calculated excluding the outlier year 2017. Please note: the panels only show statistics for the pond-time series where the pond occurred in more than three years (our threshold of detection). No ponds fulfilled that condition at the “low” site.

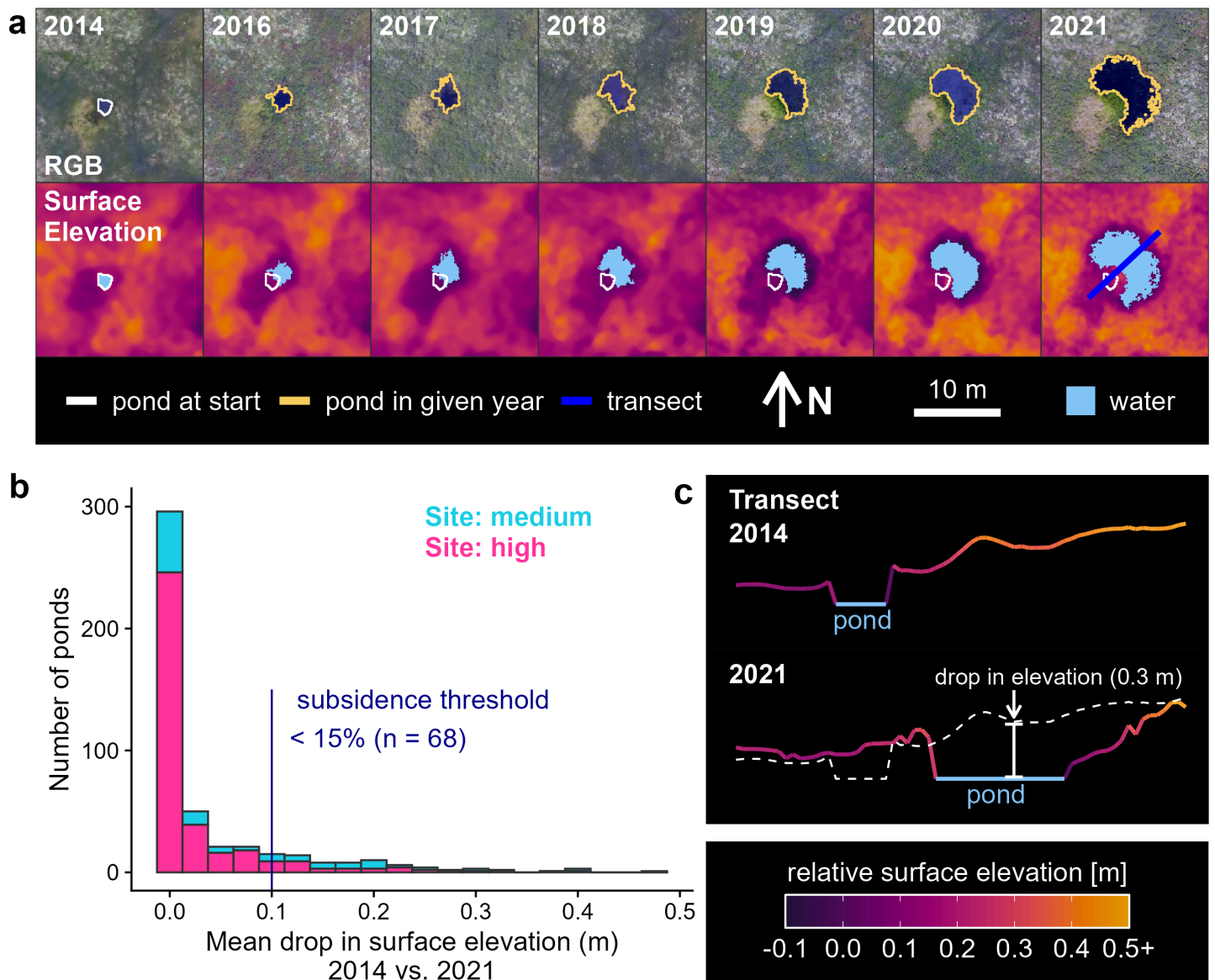


Figure 3

Ponds were short-lived and fluctuated strongly in size. The majority of ponds occurred for less than six years in the time-series and individual ponds showed a high variability in their surface area across the time-series. (a) Distribution of the number of years each pond was detected in the time-series. The dark blue line indicates the proportional split between those ponds present less than six years and those present “continuously” for six or more years (allowing for one year of failed detection). (b) Distribution of the coefficient of variation (CV) of the pond’s area across each time-series. The dark blue line indicates the mean CV value for all ponds. CV values were calculated excluding the outlier year 2017. Please note: the panels only show statistics for the pond-time series where the pond occurred in more than three years (our threshold of detection). No ponds fulfilled that condition at the “low” site.

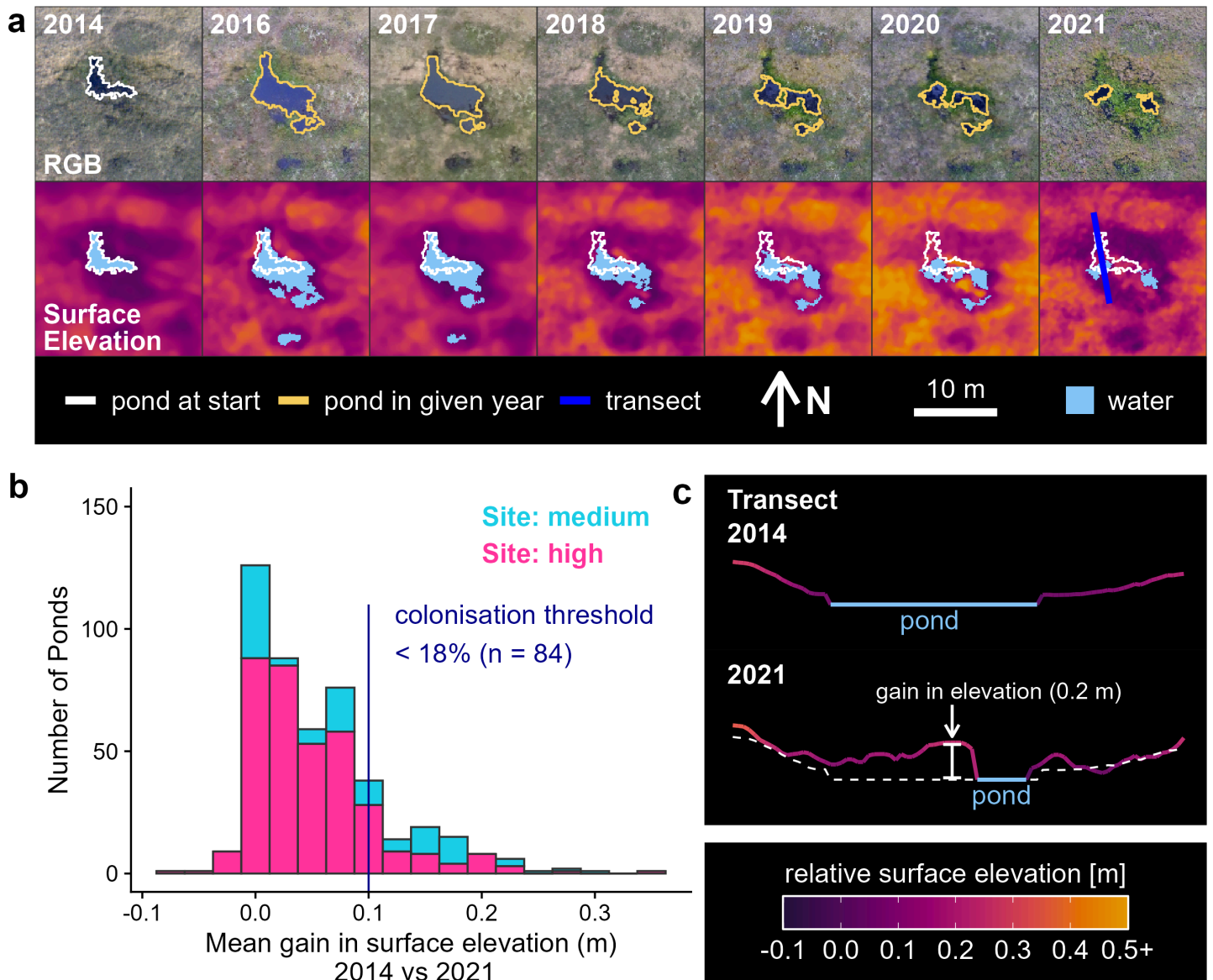


Figure 4

Vegetation colonisation contributes to shrinking or complete infilling of ponds. Example for a pond ("cbh_049") that showed clear signs of vegetation colonisation in the pond area between 2014 and 2021, observed as a gain in elevation in the drone-derived digital surface model (DSM). Upper row: true-colour (RGB) images with the outline of the pond (white / orange) in each year based on our classifications. Lower row: relative surface elevation derived from the DSM standardised to the minimum of the absolute elevation of the area shown; surface water extent in the given year (blue) and pond outline at the start of the time-series (white). (b) Distribution of the mean gain in surface elevation for the land area gained (2014 vs 2021) for all pond time-series. (c) Surface height profiles of the transect in (a) for the years 2014 and 2021, showing the gain in surface elevation due to the vegetation colonisation of pond "cbh_049" between the two years. The mean gain in surface elevation for the land area gained in pond "cbh_049" was 0.15 m.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [AssmannetalTundraPondsSupplementaryMaterials.pdf](#)